

Supporting Information

CO oxidation on IrO₂(110) surfaces

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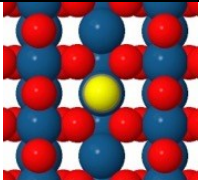
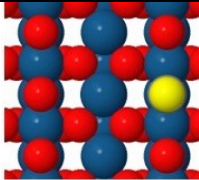
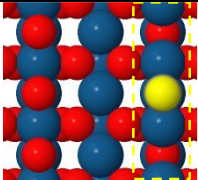
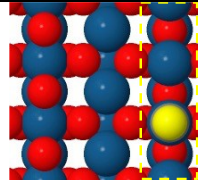
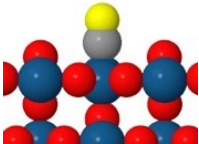
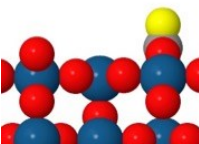
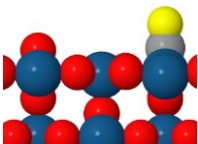
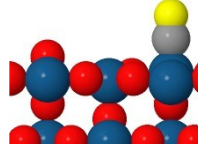
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S1. Estimating absolute CO coverages

The quantity of CO desorbing during TPRS was estimated by scaling the areas under CO TPRS traces with that determined from TPD traces acquired from CO-covered Ru(0001), and assuming a coverage of 0.68 molecules per Ru(0001) unit cell for CO adsorbed to saturation on Ru(0001) at 300 K [1]. Using areal site densities of Ir_{cus} atoms on IrO₂(110) and Ru surface atoms of Ru(0001), we obtained a multiplicative scaling factor to convert integrated CO TPD spectra to absolute CO desorption yields in units of ML. This scaling factor is multiplied by a relative sensitivity factor to estimate CO₂ desorption yields in units of ML, where the relative sensitivity factor accounts for differences in mass spectrometer sensitivity toward CO ($m/z = 28$) vs. CO₂ ($m/z = 44$) [2; 3]. Our estimates of CO TPRS yields account for electron-impact fragmentation of CO₂ to CO in the ionizer of the mass spectrometer.

S2. Comparison of CO binding energies from DFT-PBE and HSE06 calculations

Table S1. Comparison of DFT-PBE and HSE06 functional calculation in various configuration of CO adsorption. Four different configuration of CO _t , CO _{br} on stoichiometric IrO ₂ (110) (s-IrO ₂ (110)) and CO _{br,sym} and CO _{br,atop} on reduced IrO ₂ (110) (r-IrO ₂ (110)) are calculated on both [4 × 2] and [1 × 1] unit cell using PBE, HSE06 single point (sp) calculation and HSE06 full relaxation calculation. Oxygen of adsorbed CO is colored yellow and yellow dashed line indicates the reduced (O _{br,v}) row on the surface.					
		s-IrO ₂ (110)		r-IrO ₂ (110)	
		CO _t	CO _{br,sym}	CO _{br,sym}	CO _{br,atop}
					
					
[4 × 2]	PBE	223	212	206	237
	HSE06 sp	216	168	124	216
[1 × 1]	PBE	202	198	198	165
	HSE06 full	209	167	167	146

In our previous investigation of CO adsorption on PdO(101), we found that PBE functional overestimated the CO binding energies, while the HSE06 functional provided results that matched better with experimental data [4]. Similarly, we calculated the CO binding on IrO₂(110) using both PBE and HSE06 functionals. We employed both 4 × 2 unit cell and 1 × 1 unit cell for HSE06 functional calculations, using single-point calculations and full relaxation calculations, respectively. For 1 × 1 unit cell, we note that CO_{br} and CO_{br,sym} are the same configuration due to the equivalent number of O_{br,v} sites.

Table S1 shows the adsorption energies for different configurations of CO on s-IrO₂(110) and r-IrO₂(110). For the 4 × 2 unit cell, all CO adsorption configurations on both s-IrO₂(110) and r-IrO₂(110) exhibit strong binding of over 200 kJ/mol using PBE functionals. The HSE06 single point calculation, based on the PBE calculation, shows a slight decrease in energy for CO_t on s-

IrO₂(110), but a more significant decrease for the CO_{br,sym} configuration on both s-IrO₂(110) and r-IrO₂(110). The energies decrease to 168 kJ/mol and 124 kJ/mol, from 212 kJ/mol and 206 kJ/mol, respectively. This indicates that the CO_{br,sym} adsorption is more affected by the HSE06 single-point calculation. However, CO_{br,atop} configuration is less affected, similar to CO_t on s-IrO₂(110), still showing strong binding energy of over 200 kJ/mol.

We also investigated the 1×1 unit cell to perform the HSE06 full relaxation. For the 1×1 unit cell, PBE functional shows strong adsorption for both CO_t (202 kJ/mol) and CO_{br,sym} (198 kJ/mol) and weaker binding energy for the CO_{br,atop} (165 kJ/mol) configuration. The full relaxation HSE06 functional then shows an increased adsorption energy for CO_t (209 kJ/mol) and decreased adsorption energy for CO_{br,sym} (167 kJ/mol) and CO_{br,atop} (146 kJ/mol). The HSE06 full relaxation decreased the adsorption energy for CO_{br,sym} by ~ 30 kJ/mol and CO_{br,atop} by ~ 20 kJ/mol, while slightly increasing the CO_t configuration by ~ 7 kJ/mol. Overall, the HSE06 calculation effectively reduces the adsorption energies, compensating for the overestimation by the PBE functional.

S3. CO and CO₂ TPRS from s-IrO₂(110)

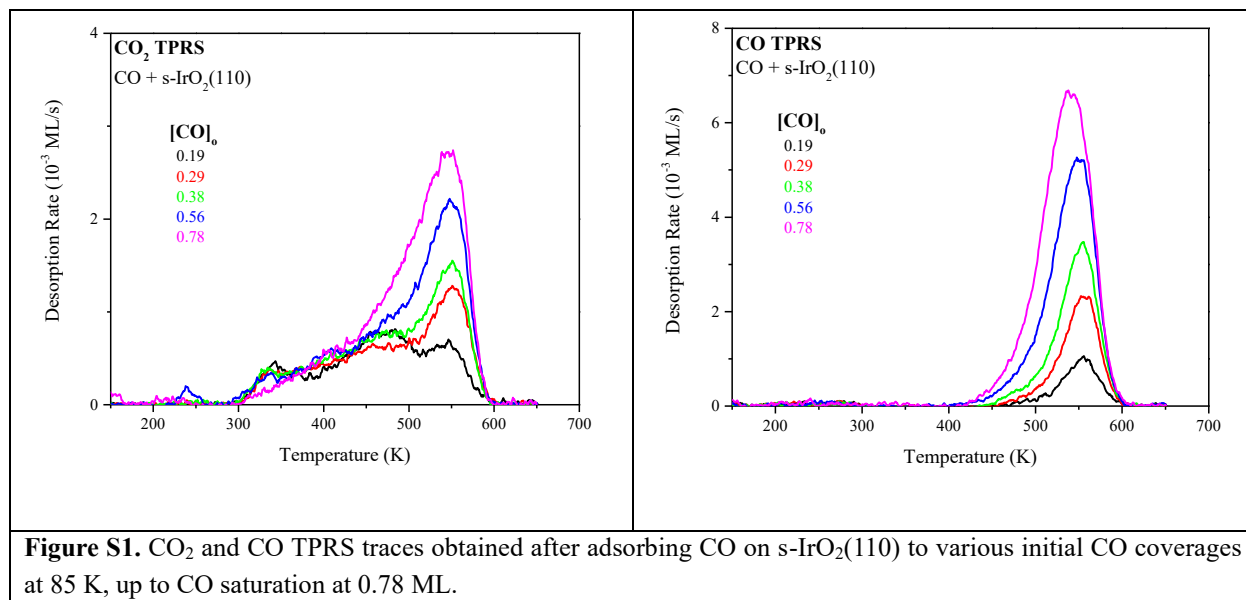
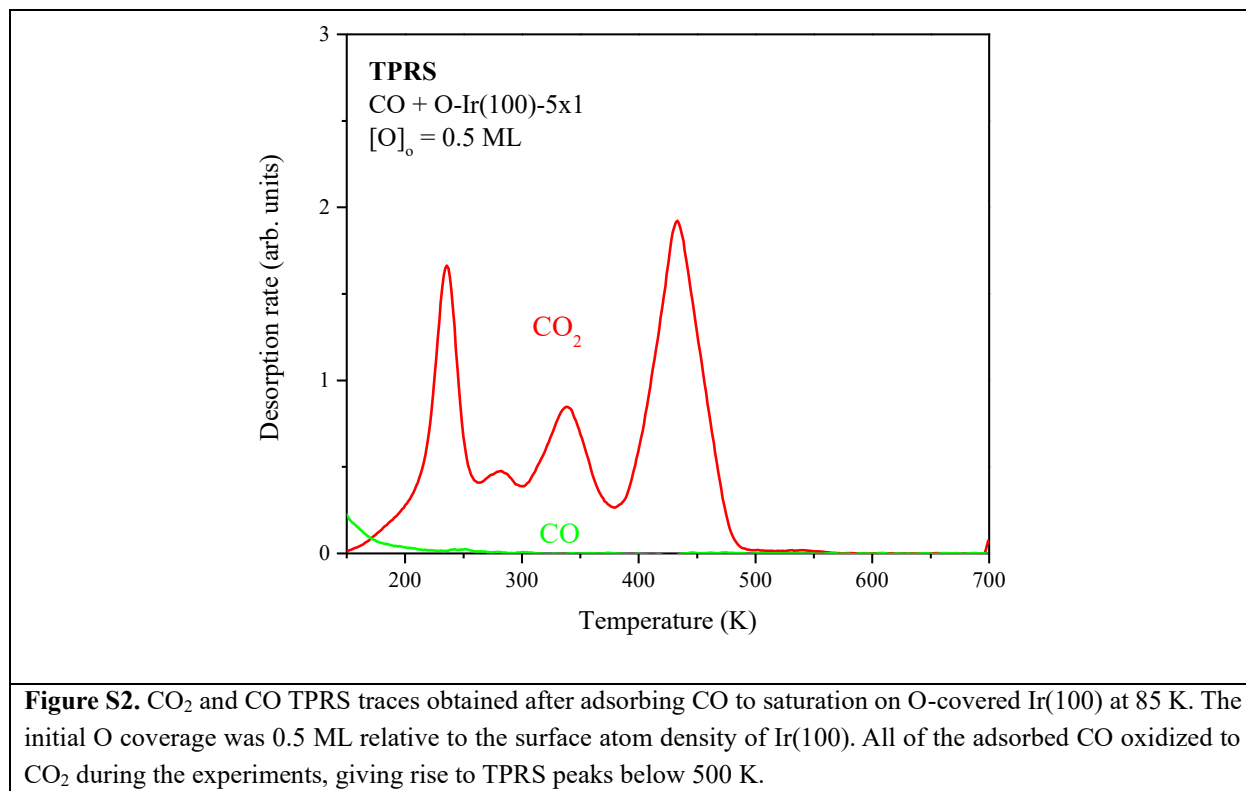
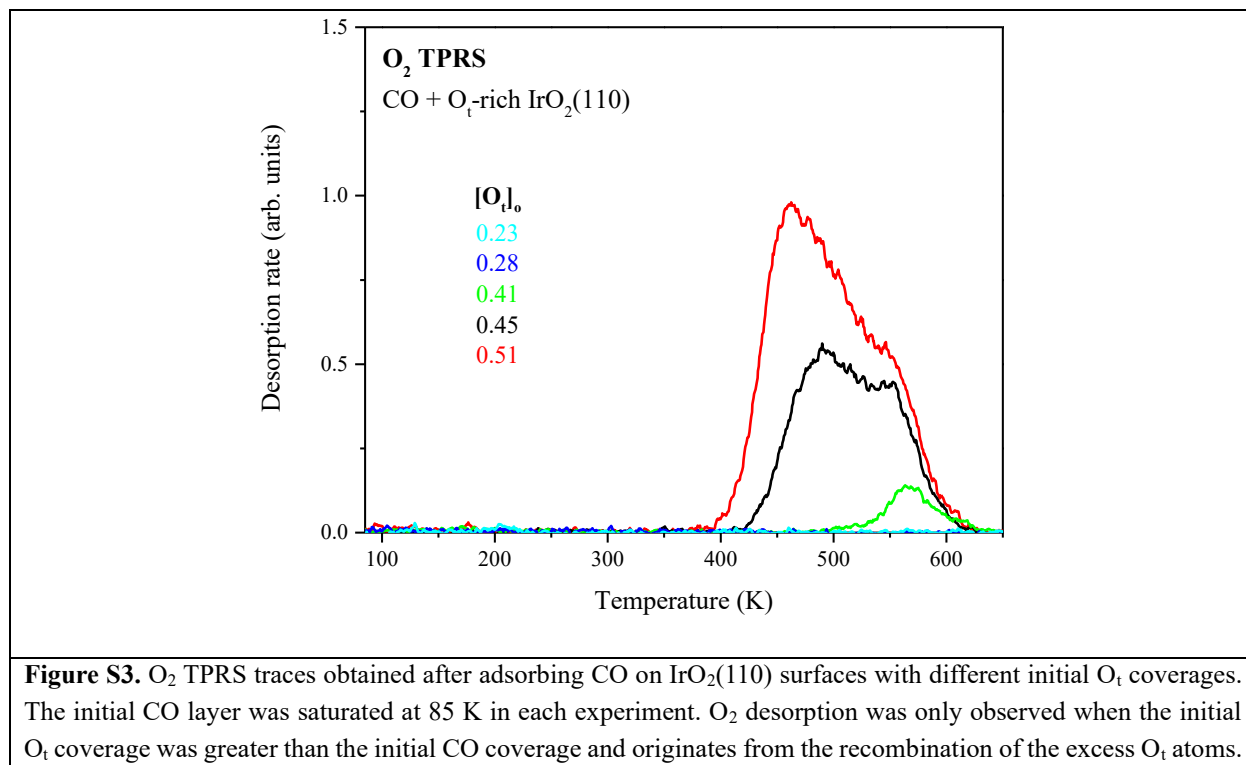


Figure S1. CO₂ and CO TPRS traces obtained after adsorbing CO on s-IrO₂(110) to various initial CO coverages at 85 K, up to CO saturation at 0.78 ML.

S4. CO and CO₂ TPRS from O-Ir(100)



S5. O₂ TPRS during CO oxidation on O_t-rich IrO₂(110)



S6. Binding of CO as a function of coverage on IrO₂(110)

We define the average adsorption energy ($E_{ads,avg}$) and differential adsorption energy ($E_{ads,diff}$) as follows:

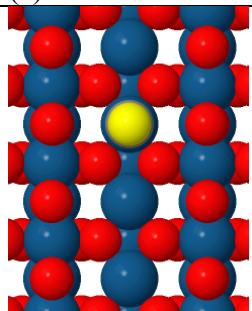
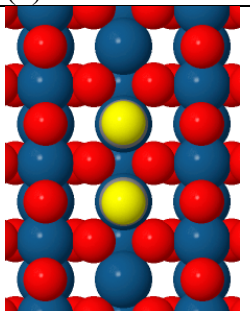
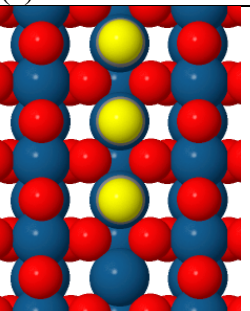
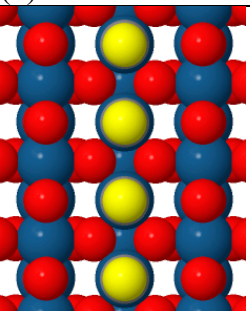
$$E_{ads,avg}(N) = \frac{E_{slab} + N * E_{CO(g)} - E_{N CO^*/slab}}{N}$$
$$E_{ads,diff}(N_2) = \frac{E_{N_1 CO^*/slab} + (N_2 - N_1) \times E_{CO(g)} - E_{N_2 CO^*/slab}}{(N_2 - N_1)}$$

Here, E_{slab} is the energy of IrO₂(110) slab and $E_{CO(g)}$ is the energy of CO gas. N is the number of adsorbed CO molecules, and $E_{N CO^*/slab}$ is the energy of system that N CO molecules adsorbed on the slab.

Figure S4 shows both $E_{ads,avg}$ and $E_{ads,diff}$ for CO adsorption on s-IrO₂(110). Both energies indicate strong CO binding across coverages from 0.25 ML to 1 ML, with negligible changes of ~ 5 kJ/mol of $E_{ads,avg}$ and ~ 10 kJ/mol of $E_{ads,diff}$. This suggests that CO is strongly adsorbed with thermodynamical favorability, with negligible repulsion between adsorbed CO molecules. We also investigated CO adsorption energy on reduced IrO₂(110) (r-IrO₂(110)), where one oxygen bridge row is vacant. Here, CO can be adsorbed on the O_{br,v} site as CO_{br,sym} (Figure S5) and on the Ir atom between O_{br,v} sites as CO_{br,atop} (Figure S6). The CO_{br,sym} configuration shows small repulsion between adsorbed CO molecules. Initially, at a coverage of 0.25 ML, the $E_{ads,avg}$ for CO_{br,sym} is calculated to be 206 kJ/mol. As the coverage increases to 1 ML, the $E_{ads,avg}$ decreases to 186 kJ/mol. This moderate decrease indicates some repulsive interaction between the adsorbed CO_{br,sym} species. Similarly, the $E_{ads,diff}$ from 0.75 ML to 1 ML of CO_{br,sym} is calculated as 144 kJ/mol. This suggests more repulsion for the CO_{br,sym} configuration on r-IrO₂(110) compared to the CO_t

configuration on s-IrO₂(110), though it remains thermodynamically favorable at full CO coverage. Additionally, the small repulsion between CO_{br,sym} configurations causes fluctuations in adsorption energies (from 0.25 ML to 0.5 ML, the energies decrease, but from 0.5 ML to 0.75 ML, the energies increase). This is due to the proximity of adsorption sites influencing each other through repulsion.

The CO_{br,atop} configuration shows the strongest repulsion among the CO adsorption species studied on IrO₂(110) surface. Initially, at a coverage of 0.25 ML of CO, the $E_{ads,avg}$ is calculated to be 237 kJ/mol. As the coverage increases to 1 ML, $E_{ads,avg}$ decreases significantly to 168 kJ/mol. This decrease of around 70 kJ/mol in adsorption energy indicates strong repulsive interaction between adsorbed CO_{br,atop} species. Similarly, the $E_{ads,diff}$ also shows a sharp decrease to 61 kJ/mol at 1 ML. This significant reduction in $E_{ads,diff}$ further confirms the increasing repulsion between CO_{br,atop} species as the coverage approaches 1 ML. Despite this strong repulsion, the adsorption of CO_{br,atop} remains thermodynamically favored, as indicated by the positive value of adsorption energies. This suggests that even though the repulsion increases with higher coverage, the overall process of CO_{br,atop} adsorption on r-IrO₂(110) is still thermodynamically favorable.

	(a)	(b)	(c)	(d)
				
$E_{ads,avg}$	223	221	219	217
$E_{ads,diff}$	223	219	214	211
Figure S4. CO _t adsorption energies (average and differential) on s-IrO ₂ (110), (a) 0.25 ML, (b) 0.5 ML, (c) 0.75 ML, and (d) 1 ML. O atoms of CO _t is colored yellow.				

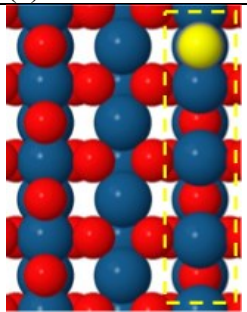
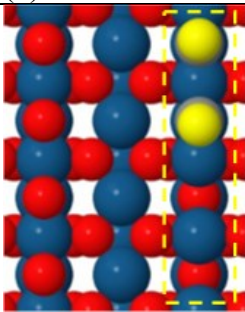
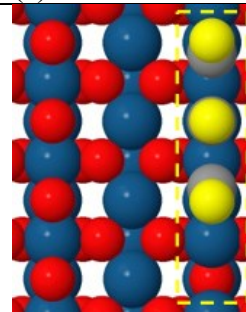
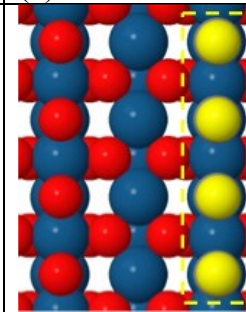
	(a)	(b)	(c)	(d)
				
$E_{ads,avg}$	206	194	200	186
$E_{ads,diff}$	206	182	213	144

Figure S5. $\text{CO}_{\text{br,sym}}$ adsorption energies (average and differential) on reduced $\text{IrO}_2(110)$ (r- $\text{IrO}_2(110)$), (a) 0.25 ML, (b) 0.5 ML, (c) 0.75 ML, and (d) 1 ML. O atoms of $\text{CO}_{\text{br,sym}}$ is colored yellow, and yellow dashed line indicate the reduced $\text{O}_{\text{br,v}}$ row.

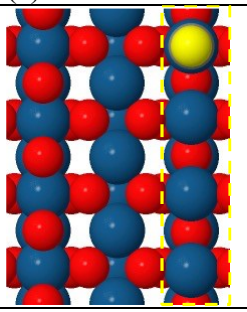
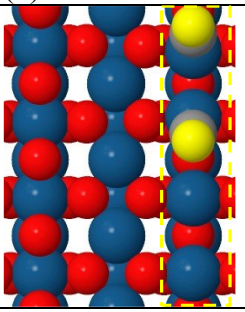
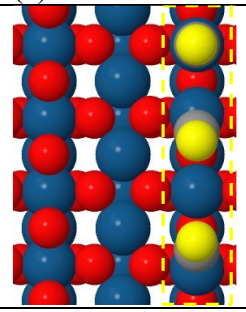
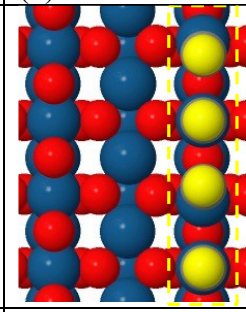
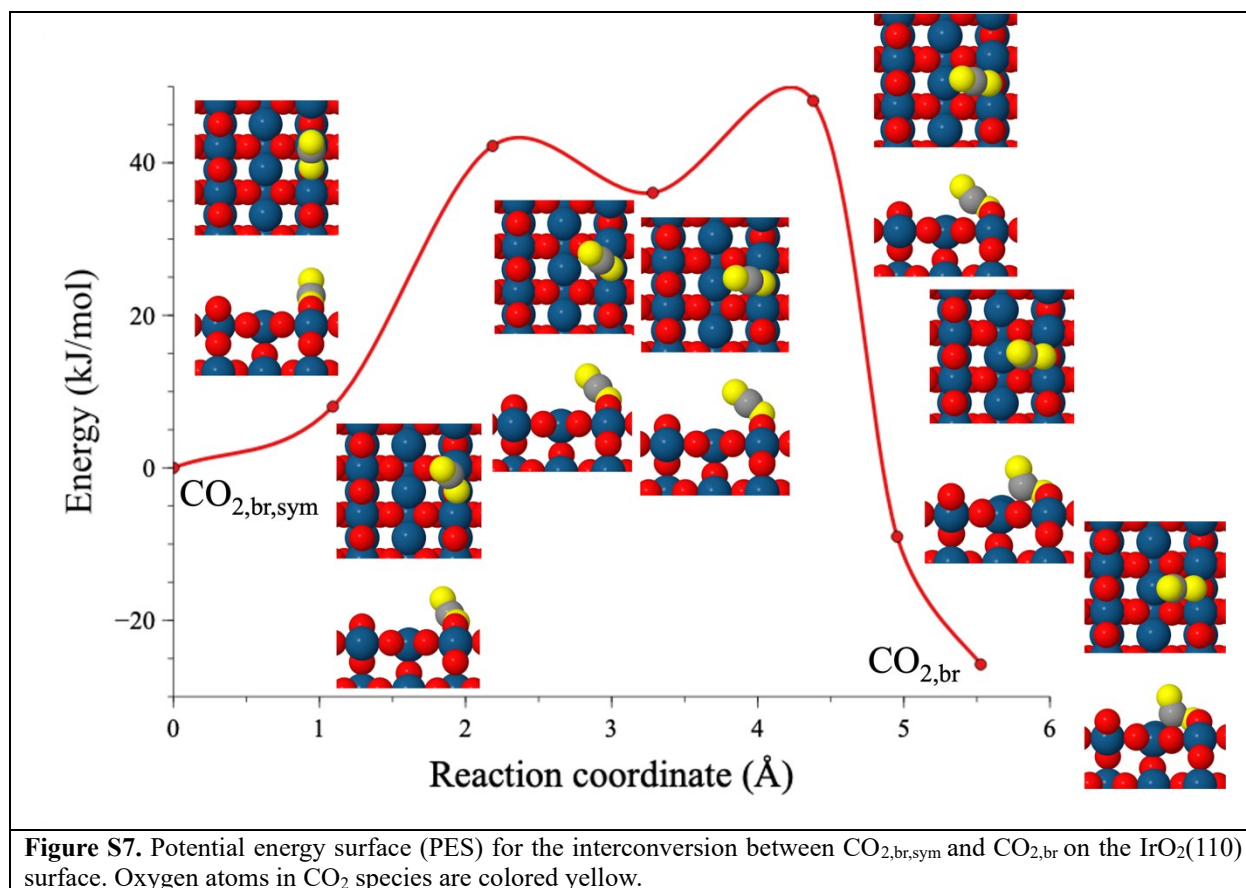
	(a)	(b)	(c)	(d)
				
$E_{ads,avg}$	237	229	204	168
$E_{ads,diff}$	237	220	154	61

Figure S6. $\text{CO}_{\text{br,atop}}$ adsorption energies (average and differential) on reduced $\text{IrO}_2(110)$ (r- $\text{IrO}_2(110)$), (a) 0.25 ML, (b) 0.5 ML, (c) 0.75 ML, and (d) 1 ML. O atoms of $\text{CO}_{\text{br,atop}}$ is colored yellow, and yellow dashed line indicate the reduced $\text{O}_{\text{br,v}}$ row.

S7. Interconversion of CO_{2,br} species on IrO₂(110) surface

We investigated various pathways for CO oxidation on IrO₂(110) surface, as shown in Figures 9 and 10 of the main manuscript. Notably, the CO oxidation process on the IrO₂(110)-1*_{br} surface shows distinct behavior regarding the formed CO₂ species. As depicted in Figure 9 (b) of the main manuscript, the reaction sequence initiates with the adsorption of CO as a CO_t species on the surface. This CO_t species subsequently migrates, filling a vacancy and transforming into a CO_{br,sym} species. The CO_{br,sym} species then reacts with a surface oxygen atom, forming a CO_{2,br} species. In this process, the CO_{br,sym} reacts with adjacent surface oxygen atom to form CO_{2,br,sym} species (we denote CO₂ formed by CO_{br,sym} and O_{br} as CO_{2,br,sym} to distinguish it from the CO_{2,br} generated from the reaction of CO_t and O_{br}). The CO_{2,br,sym} species can undergo a rotational interconversion on the surface, resulting in the formation of CO_{2,br}, a step omitted in Figure 9 (b).

Figure S7 shows the potential energy surface (PES) for the interconversion between CO_{2,br,sym} and CO_{2,br} on the surface. The pathway involves multiple transition states, implying multiple elementary reactions, with the energy indicating an exothermic process. The CO₂ species at higher energy points, around transition states, are also depicted, showing the linear structure and rotational dynamics of the CO_{2,br,sym} and CO_{2,br} species. DFT vibrational frequency analysis reveals that the linear CO₂ species formed during this process exhibit imaginary frequencies, indicating that these species are transient and involved in this process as unstable states. Therefore, we approximated the forward barrier for this conversion process to be around 40 kJ/mol, while the reverse barrier is around 70 kJ/mol. These values were incorporated into our microkinetic model. The microkinetic model, using these approximated barriers, aligns well with experimental results. It is notable that this specific DFT calculations was performed with a convergence criterion of 0.05 eV/Å, in contrast to the 0.03 eV/Å criterion used for other calculations in this study.



S8. Rate equations and kinetic parameters used in the microkinetic model

Table S2. Elementary reaction steps, energy barriers (E_f , E_r) and pre-factors (A_f , A_r) for the forward and reverse reactions, and the rate equations used in the microkinetic model for CO oxidation on IrO₂(110). The energies shown in parentheses were determined from single-point HSE calculations, while other energies were computed using PBE. Energies are given in units of kJ/mol and pre-factors in units of s⁻¹. Elementary rate equations were used in all cases, where k_j and [*species i*] represent the rate coefficient for reaction j and the surface coverage of species i . The quantities $P_{O_{br}}$, $P_{*_{br}}$ and $P_{2*_{br}}$ represent the probabilities that CO is adsorbed next to an O_{br} atom, a single *_{br} site or a pair of *_{br} sites, respectively. Equations for these configuration probabilities were determined using the mean-field approximation as described below. ^(a)The energy barrier for forward reaction (3) was set equal to 139 kJ/mol, or 76% of the PBE value, to optimize agreement between the simulated and experimental TPRS traces.

	Reaction step	E_f	E_r	A_f	A_r	Forward rate	Reverse rate
(1)	$\text{CO}_t \rightarrow \text{CO(g)} + *_{\text{cus}}$	223 (216)	--	5×10^{15}	--	$k_1[\text{CO}_t]$	--
(2)	$\text{CO}_t + \text{O}_{\text{br}} \rightleftharpoons \text{CO}_{2,\text{br}} + *_{\text{cus}}$	95	0	5×10^{12}	5×10^{12}	$k_{2f}[\text{CO}_t]P_{O_{br}}$	$k_{2r}[\text{CO}_{2,\text{br}}][*_{\text{cus}}]$
(3)	$\text{CO}_t + *_{\text{br}} \rightleftharpoons \text{CO}_{\text{br,sym}} + *_{\text{cus}}$	139 ^(a)	172	5×10^{12}	5×10^{12}	$k_{3f}[\text{CO}_t]P_{*_{br}}$	$k_{3r}[\text{CO}_{\text{br,sym}}][*_{\text{cus}}]$
(4)	$\text{CO}_t + 2*_{\text{br}} \rightleftharpoons \text{CO}_{\text{br,atop}} + *_{\text{cus}}$	182	198	5×10^{12}	5×10^{12}	$k_{4f}[\text{CO}_t]P_{2*_{br}}$	$k_{4r}[\text{CO}_{\text{br,atop}}][*_{\text{cus}}]$
(5)	$\text{CO}_{\text{br,sym}} \rightarrow \text{CO(g)} + *_{\text{br}}$	212 (168)	--	5×10^{15}	--	$k_5[\text{CO}_{\text{br,sym}}]$	--
(6)	$\text{CO}_{\text{br,atop}} \rightarrow \text{CO(g)} + 2*_{\text{br}}$	237 (216)	--	5×10^{15}	--	$k_6[\text{CO}_{\text{br,atop}}]$	--
(7)	$\text{CO}_{\text{br,sym}} + \text{O}_{\text{br}} \rightleftharpoons \text{CO}_{2,\text{br,sym}} + *_{\text{br}}$	148	40	5×10^{12}	5×10^{12}	$k_{7f}[\text{CO}_{\text{br,sym}}][\text{O}_{\text{br}}]$	$k_{7r}[\text{CO}_{2,\text{br}}][*_{\text{br}}]$
(8)	$\text{CO}_{2,\text{br,sym}} \rightleftharpoons \text{CO}_{2,\text{br}}$	40	70	5×10^{12}	5×10^{12}	$k_{8f}[\text{CO}_{2,\text{br,sym}}]$	$k_{8r}[\text{CO}_{2,\text{br}}]$
(9)	$\text{CO}_{2,\text{br,sym}} \rightarrow \text{CO}_2(\text{g}) + *_{\text{br}}$	38	--	5×10^{13}	--	$k_9[\text{CO}_{2,\text{br,sym}}]$	--
(10)	$\text{CO}_{2,\text{br}} \rightarrow \text{CO}_2(\text{g}) + *_{\text{br}}$	68	--	5×10^{13}	--	$k_{10}[\text{CO}_{2,\text{br}}]$	--
(11)	$\text{CO}_t + \text{O}_t \rightleftharpoons \text{CO}_{2,\text{t}} + *_{\text{cus}}$	62	82	5×10^{12}	5×10^{12}	$k_{11f}[\text{CO}_t][\text{O}_t]$	$k_{11r}[\text{CO}_{2,\text{t}}][*_{\text{cus}}]$
(12)	$\text{CO}_{2,\text{t}} \rightarrow \text{CO}_2(\text{g}) + *_{\text{cus}}$	38	--	5×10^{13}	--	$k_{12}[\text{CO}_{2,\text{t}}]$	--
(13)	$2\text{O}_t \rightarrow \text{O}_2(\text{g}) + 2*_{\text{cus}}$	182	--	5×10^{13}	--	$k_{13}[\text{O}_t]^2$	--

The microkinetic model developed for simulating CO oxidation on IrO₂(110) surfaces includes elementary rate equations, and uses energy barriers mainly determined from DFT calculations. Specifically, the optimum agreement with the experimental TPRS data, as shown in the main manuscript, was obtained by setting the CO desorption barriers equal to the CO binding energies determined from single-point HSE06 calculations, and setting the barrier for CO_t to CO_{br,sym} conversion equal to 139 kJ/mol, which is 76% of the PBE value. All other energy barriers used in the simulations were taken directly from DFT-PBE calculations. The pre-factors used in the model

were set to physically reasonable values and adjusted to refine agreement between the simulated and experimental TPRS curves. The model assumes that CO_t molecules can react with an O_{br} atom to generate an CO_{2,br} species, or convert to a CO_{br,sym} or CO_{br,atop} species by moving to *_{br} sites. We assume that CO_{br,atop} formation requires a pair of neighboring *_{br} sites, and that the rate of this reaction is proportional to the probability that a pair of *_{br} sites is present on a bridging row, P_{2*br} . Since each CO_t molecule is adsorbed next to two bridging rows, we determined an equation for P_{2*br} by considering the possible configurations of 4 bridging sites, comprised of pairs of sites located on adjacent rows. For this configuration, the mean-field probability of a *_{br} site pair is given by the equation,

$$P_{2*br} = 2[*_{br}]^2(1 - [*_{br}])^2 + 4[*_{br}]^3(1 - [*_{br}]) + [*_{br}]^4.$$

The coverage of *_{br} sites is set equal to $[*_{br}] = 1 - [O_{br}] - [CO_{br,sym}] - 2[CO_{br,atop}] - [CO_{2,br}]$, where the factor of 2 assumes that each CO_{br,atop} species effectively occupies two *_{br} sites. For consistency, we assumed that the rates of CO_t oxidation by an O_{br} atom and CO_t conversion to a CO_{br,sym} species are proportional to the mean-field probabilities that a CO_t molecule is next to an O_{br} atom (P_{Obr}) or a *_{br} site (P_{*br}) within the 4-site configuration as given by the following equations,

$$P_{Obr} = [O_{br}]^4 + 4[O_{br}]^3(1 - [O_{br}]) + 5[O_{br}]^2(1 - [O_{br}])^2 + 2[O_{br}](1 - [O_{br}])^3$$

and,

$$P_{*br} = 2[*_{br}](1 - [*_{br}])^3 + 5[*_{br}]^2(1 - [*_{br}])^2 + 4[*_{br}]^3(1 - [*_{br}]) + [*_{br}]^4$$

S9. TPRS simulations

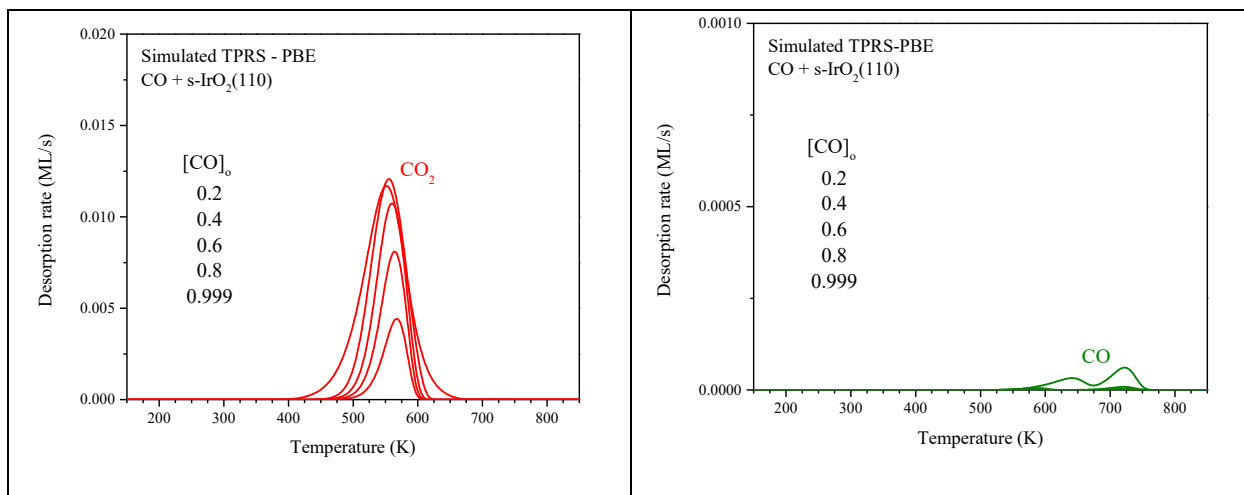


Figure S8. Simulated CO₂ and CO TPRS traces after adsorbing CO on s-IrO₂(110) to various initial CO coverages. All energy barriers were determined from DFT-PBE calculations (Table S1). The fractional CO₂ yield remains greater than 99% as the initial CO coverage is increased from 0.2 ML to nearly 1 ML, exceeding the experimental values.

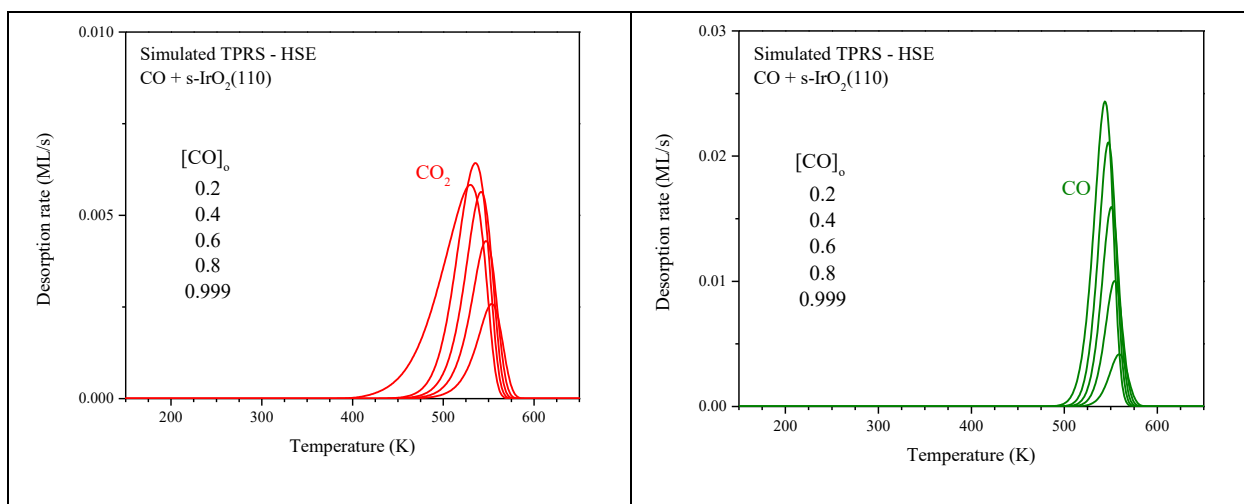


Figure S9. Simulated CO₂ and CO TPRS traces after adsorbing CO on s-IrO₂(110) to various initial CO coverages. The simulations used CO binding energies (desorption barriers) determined from single-point HSE calculations, and set the barrier for CO_t to CO_{br,sym} conversion equal to 139 kJ/mol. All other energy barriers were determined from DFT-PBE calculations (Section S8). The fractional CO₂ yield decreases from 47% to 35% as the initial CO coverage is increased from 0.2 ML to nearly 1 ML, and CO₂ and CO desorb at similar temperatures between ~530 and 560 K, in good agreement with experimental observations.

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